

Worksheet Part W5: the $p = 5$ table

accompanying *Mild p -class tower groups of imaginary quadratic fields*

References of the form Proposition 2.12, §4.2, or (P. n) refer to statements, sections, and numbered equations of the paper.

Part W5: The $p = 5$ table

The 204 imaginary quadratic fields with 5-class rank three and $|D_K| < 2^{30}$. Table 1 lists the 61 fields of the first block $|D_K| < 2^{28}$ in full detail; its columns record the class group, the number of $\mathbb{F}_5$ -rational points of the norm-degeneracy scheme Σ_D (Definition 3.17 of the paper) at which $\text{rk } D_x = 1$, the number of these which are transverse, and the smallest degree of an extension of $\mathbb{F}_5$ over which a transverse point exists. The defining polynomials of the extensions used for the fields without a rational transverse point, the coordinates of all transverse points, and the corresponding transversality matrices are recorded for every field of the census in `examples/p5/transverse-rank-one/` (`report.txt` human-readable, `certificates.json` machine-readable); the finite-field calculations underlying both tables are re-run deterministically by `tools/transverse\_rank\_one.py`.

	D_K	$\text{Cl}(K)$	rank-one over $\mathbb{F}_5$	transverse over $\mathbb{F}_5$	least transverse degree
1	-11203620	[10, 10, 10]	2	1	1
2	-18397407	[40, 10, 5]	3	3	1
3	-27960639	[40, 10, 10]	2	2	1
4	-35663739	[30, 10, 5]	0	0	6
5	-51213139	[75, 5, 5]	1	0	4
6	-54319112	[60, 10, 5]	4	2	1
7	-61040707	[20, 10, 5]	2	1	1
8	-65818135	[30, 10, 10]	3	2	1
9	-75949255	[40, 20, 5]	2	2	1
10	-77778287	[120, 10, 5]	3	3	1
11	-89017304	[170, 10, 5]	2	2	1
12	-89218664	[310, 5, 5]	1	1	1
13	-90903207	[110, 10, 5]	2	2	1
14	-93121640	[90, 10, 10]	1	1	1
15	-104545864	[60, 10, 5]	1	1	1
16	-106660295	[110, 10, 10]	2	2	1
17	-109909943	[100, 10, 10]	1	1	1
18	-123482119	[270, 5, 5]	4	4	1
19	-123779560	[10, 10, 10, 2]	2	1	1
20	-126740891	[195, 5, 5]	0	0	6
21	-129063459	[60, 10, 5]	1	1	1
22	-130370807	[100, 10, 10]	0	0	2
23	-134005207	[100, 10, 5]	3	2	1
24	-145367147	[125, 5, 5]	2	2	1
25	-150535579	[140, 5, 5]	2	2	1
26	-150679519	[150, 10, 5]	0	0	6
27	-150752699	[160, 5, 5]	2	1	1

	D_K	$\text{Cl}(K)$	rank-one over $\mathbb{F}_5$	transverse over $\mathbb{F}_5$	least transverse degree
28	-156145931	[165, 5, 5]	3	3	1
29	-164995035	[60, 10, 5]	1	0	2
30	-165570488	[60, 20, 5]	1	1	1
31	-167094823	[60, 10, 10]	1	1	1
32	-176386616	[180, 10, 5]	2	1	1
33	-181738571	[100, 10, 5]	2	1	1
34	-182240771	[225, 5, 5]	3	3	1
35	-186243704	[40, 20, 10]	0	0	3
36	-187160831	[60, 20, 10]	2	1	1
37	-190572839	[630, 5, 5]	2	2	1
38	-194080808	[70, 10, 10]	1	1	1
39	-199674799	[340, 5, 5]	1	1	1
40	-202331347	[30, 10, 5]	1	1	1
41	-203145767	[60, 10, 10, 2]	2	1	1
42	-203437103	[210, 10, 5]	3	2	1
43	-204909944	[80, 10, 10]	1	1	1
44	-205996583	[100, 10, 10]	1	1	1
45	-207666763	[50, 10, 5]	1	1	1
46	-209149784	[180, 10, 5]	3	2	1
47	-212229543	[60, 10, 10]	1	0	3
48	-216770467	[95, 5, 5]	1	1	1
49	-225826216	[20, 10, 10, 2]	1	1	1
50	-226373428	[30, 10, 10]	3	3	1
51	-231112315	[20, 10, 10]	2	2	1
52	-239452895	[160, 10, 10]	1	1	1
53	-242963687	[370, 5, 5]	0	0	6
54	-246238555	[70, 10, 5]	1	1	1
55	-247127879	[160, 10, 10]	1	1	1
56	-248046051	[60, 10, 5]	1	1	1
57	-249407767	[60, 10, 10]	0	0	6
58	-252735992	[280, 5, 5]	5	4	1
59	-253425979	[135, 5, 5]	0	0	3
60	-256232447	[420, 5, 5]	1	1	1
61	-261286872	[10, 10, 10, 2, 2]	1	1	1

Table 2 lists the 142 mild fields with $2^{28} \leq |D_K| < 2^{30}$, with the class group and the least degree of an extension of $\mathbb{F}_5$ carrying a transverse point. The one remaining field of the range, $D_K = -781922404$ with $\text{Cl}(K) \simeq [120, 15, 5]$, is not listed: its norm-degeneracy scheme consists of a single non-reduced closed point of degree two, so no transverse point exists over any extension.

	D_K	$\text{Cl}(K)$	least transverse degree
1	-272394484	[150, 10, 5]	6
2	-280557591	[150, 10, 10]	1
3	-282198803	[120, 10, 5]	1
4	-285889891	[60, 10, 5]	1
5	-294897647	[100, 10, 10]	6

	D_K	$\text{Cl}(K)$	least transverse degree
6	-300240404	[250, 10, 5]	1
7	-308728199	[820, 5, 5]	2
8	-311069163	[60, 10, 5]	1
9	-312163691	[130, 10, 5]	1
10	-313065188	[140, 10, 5]	1
11	-330727336	[150, 5, 5]	6
12	-334692215	[80, 10, 10, 2]	2
13	-342156463	[200, 10, 5]	1
14	-343576036	[220, 5, 5]	2
15	-344438804	[300, 10, 5]	2
16	-345369167	[160, 10, 10]	1
17	-348003156	[70, 10, 10]	1
18	-351044647	[280, 5, 5]	1
19	-360803647	[310, 5, 5]	1
20	-364780660	[70, 10, 10]	1
21	-368523908	[20, 20, 10, 2]	2
22	-416349256	[40, 10, 10]	1
23	-417286132	[110, 10, 5]	1
24	-429818367	[120, 10, 10]	1
25	-433597399	[240, 10, 5]	1
26	-434084164	[300, 5, 5]	2
27	-439073371	[70, 10, 5]	1
28	-443585299	[100, 10, 5]	1
29	-444408443	[120, 10, 5]	1
30	-448376516	[110, 10, 10]	1
31	-467347512	[20, 10, 10, 2]	1
32	-469835647	[80, 20, 5]	1
33	-480094015	[350, 5, 5]	1
34	-480642151	[270, 10, 5]	4
35	-482644063	[100, 10, 10]	1
36	-498607924	[160, 10, 5]	1
37	-501070231	[775, 5, 5]	1
38	-506335479	[190, 10, 10]	1
39	-508163803	[30, 10, 10]	2
40	-509504379	[140, 10, 5]	6
41	-512578628	[80, 10, 10]	2
42	-516168299	[450, 5, 5]	2
43	-520957412	[230, 10, 5]	1
44	-522401819	[320, 5, 5]	1
45	-527734248	[80, 10, 10]	1
46	-541579031	[280, 10, 10]	1
47	-545873304	[70, 10, 10]	1
48	-558181271	[1095, 5, 5]	1
49	-568232120	[50, 10, 10, 2]	1
50	-573379016	[60, 10, 10, 2]	1
51	-578286527	[110, 10, 10, 2]	1
52	-583175288	[50, 10, 10, 2]	2
53	-587516655	[40, 10, 10, 2, 2]	1
54	-587585540	[60, 10, 10, 2]	1
55	-600487131	[80, 10, 10]	1
56	-611642824	[50, 10, 10]	1
57	-614889363	[20, 20, 10]	1

	D_K	$\text{Cl}(K)$	least transverse degree
58	-617556791	[270, 10, 10]	1
59	-618229927	[360, 5, 5]	1
60	-638094767	[730, 5, 5]	1
61	-648249064	[70, 10, 10]	1
62	-648716003	[300, 5, 5]	2
63	-652187816	[130, 10, 10]	1
64	-654932024	[330, 10, 5]	6
65	-659084579	[570, 5, 5]	2
66	-663884564	[140, 10, 10]	1
67	-671772331	[320, 5, 5]	1
68	-672002387	[130, 10, 5]	1
69	-684312527	[600, 10, 5]	4
70	-690429160	[20, 10, 10, 2]	1
71	-694030820	[130, 10, 10]	1
72	-694740327	[240, 10, 5]	6
73	-703985192	[390, 5, 5]	1
74	-716214548	[130, 10, 10]	6
75	-716545112	[310, 10, 5]	1
76	-717538411	[215, 5, 5]	1
77	-719003311	[340, 10, 5]	1
78	-720658367	[390, 10, 5]	1
79	-729506291	[520, 5, 5]	1
80	-730579195	[40, 20, 5]	1
81	-736568551	[705, 5, 5]	1
82	-747498712	[100, 10, 5]	2
83	-747906091	[210, 5, 5]	1
84	-752704467	[130, 5, 5]	1
85	-753660888	[60, 10, 10]	1
86	-758811723	[110, 10, 5]	1
87	-770289259	[275, 5, 5]	1
88	-792236011	[130, 10, 5]	1
89	-818819903	[320, 10, 5]	1
90	-822797303	[480, 10, 5]	1
91	-823763252	[200, 10, 5]	6
92	-829075271	[1490, 5, 5]	1
93	-832956936	[50, 10, 10, 2]	1
94	-834867768	[30, 10, 10, 2]	1
95	-835199035	[50, 10, 10]	1
96	-837148756	[150, 10, 5]	3
97	-839189588	[160, 10, 10]	1
98	-847946303	[1020, 5, 5]	3
99	-848798407	[375, 5, 5]	1
100	-856365643	[170, 5, 5]	1
101	-859104103	[230, 10, 5]	3
102	-860862356	[710, 5, 5]	1
103	-860938607	[390, 10, 5]	2
104	-863732452	[90, 10, 5]	1
105	-864855431	[60, 60, 10]	1
106	-865276015	[280, 10, 5]	3
107	-865288212	[60, 10, 10, 2]	1
108	-865931992	[60, 10, 10]	1
109	-873927415	[240, 10, 5]	1

	D_K	$\text{Cl}(K)$	least transverse degree
110	-876422059	[265, 5, 5]	4
111	-880112408	[270, 10, 5]	1
112	-880342436	[660, 5, 5]	1
113	-881852344	[80, 20, 5]	1
114	-882321235	[130, 10, 5]	6
115	-884383779	[180, 10, 5]	1
116	-890032871	[750, 10, 5]	1
117	-912814943	[600, 10, 5]	1
118	-930871220	[160, 10, 10]	6
119	-937521204	[60, 20, 10]	1
120	-941867380	[180, 10, 5]	1
121	-954227011	[270, 5, 5]	6
122	-964632455	[1180, 5, 5]	1
123	-967077539	[450, 5, 5]	1
124	-975828939	[270, 5, 5]	1
125	-976016443	[210, 5, 5]	1
126	-979106379	[410, 5, 5]	6
127	-996554680	[60, 10, 10]	2
128	-1007496479	[920, 10, 5]	1
129	-1008688623	[850, 5, 5]	4
130	-1011574052	[60, 20, 10]	3
131	-1014273539	[360, 5, 5]	6
132	-1016848324	[300, 5, 5]	1
133	-1018718571	[180, 10, 5]	1
134	-1019215031	[450, 15, 5]	6
135	-1030705032	[40, 10, 10, 2]	1
136	-1040930248	[120, 10, 5]	1
137	-1041244847	[470, 10, 5]	2
138	-1043658971	[550, 5, 5]	2
139	-1047634431	[1030, 5, 5]	1
140	-1064940171	[180, 10, 5]	1
141	-1066461896	[330, 10, 5]	6
142	-1069759951	[360, 10, 5]	1